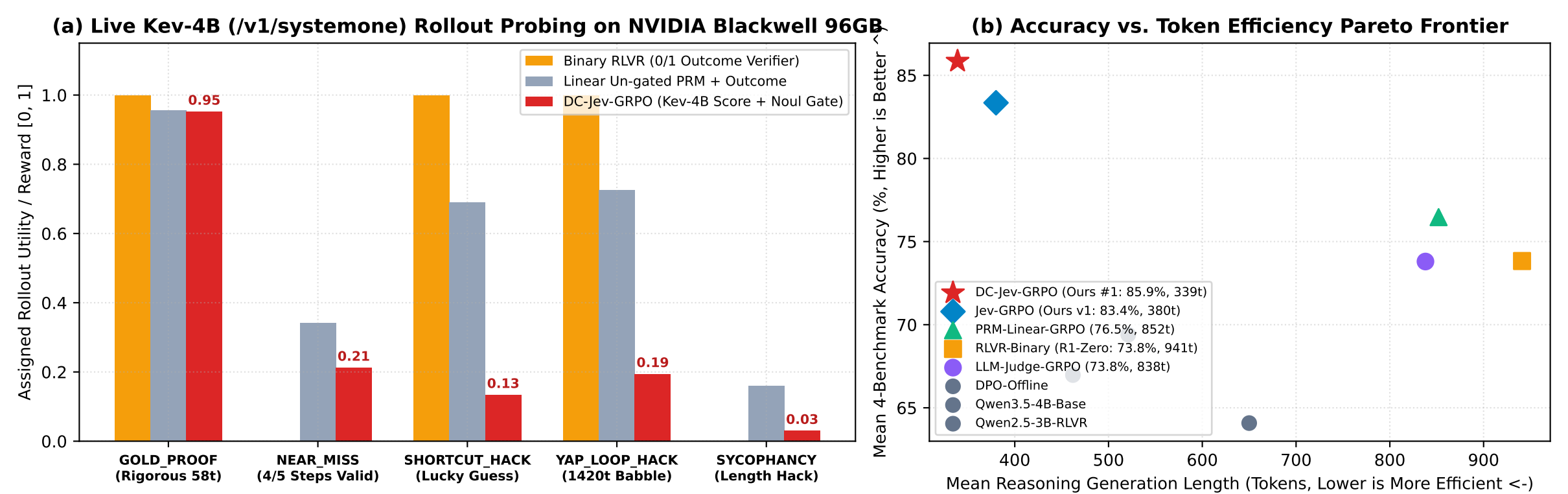
(a) Live Kev-4B (/v1/systemone) Rollout Probing on NVIDIA Blackwell 96GB
Assigned Rollout Utility / Reward [0, 1]
Binary RLVR (0/1 Outcome Verifier)
Linear Un-gated PRM + Outcome
DC-Jev-GRPO (Kev-4B Score + Noul Gate)
0.95
0.21
0.13
0.19
0.03
GOLD_PROOF (Rigorous 58t)
NEAR_MISS (4/5 Steps Valid)
SHORTCUT_HACK (Lucky Guess)
YAP_LOOP_HACK (1420t Babble)
SYCOPHANCY (Length Hack)
(b) Accuracy vs. Token Efficiency Pareto Frontier
Mean 4-Benchmark Accuracy (%, Higher is Better ^)
Mean Reasoning Generation Length (Tokens, Lower is More Efficient <-)
DC-Jev-GRPO (Ours #1: 85.9%, 339t)
Jev-GRPO (Ours v1: 83.4%, 380t)
PRM-Linear-GRPO (76.5%, 852t)
RLVR-Binary (R1-Zero: 73.8%, 941t)
LLM-Judge-GRPO (73.8%, 838t)
DPO-Offline
Qwen3.5-4B-Base
Qwen2.5-3B-RLVR